LungCT AI | CLINICAL NOISE REPORT (Model 1 (U-Net++))

Date: 2026-09-03 08:52:57
Protocol: Model 1 (U-Net++)

Case / Scan ID:	LCT-85725	**Patient Case:**	ANONYMOUS (Clinical Study)
File Name:	test_ct_slice.png	**Analysis Model:**	U-Net++ Nested Dense Segmentation (Gaussian + Poisson)
Modality:	Computed Tomography (CT)	**Report Status:**	**COMPLETED (AI-Generated)**

CT Scan Imagery & Diagnostic Overlay

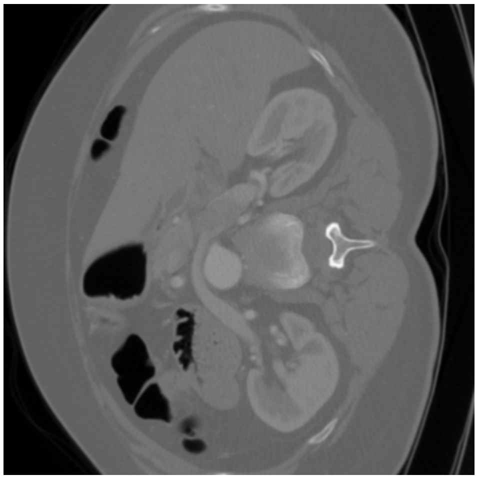

Figure 1: Original CT Scan
Input diagnostic image

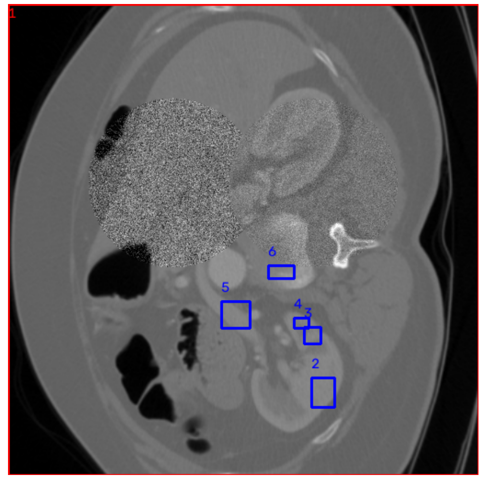

Figure 2: Annotated CT Scan (Red: Gaussian | Blue: Poisson)

Quantitative Noise Classification & Metrics (Model 1 (U-Net++))

Noise Classification	Pixel Count / Prob.	Severity / Score	Clinical Classification
Gaussian Noise (Electronic)	250,356 px	95.50%	Critical
Poisson Noise (Photon Shot)	1,523 px	0.58%	Mild
Clean Signal Region	10,265 px	3.92%	Normal / Target
TOTAL NOISE (Model 1)	**251,879 px**	**96.08%**	**Critical**

Clinical Interpretation & Denoising Recommendations

Model Assessment (Model 1 (U-Net++)): Elevated noise artifacts detected (96.1%). Structural detail is noticeably compromised by noise patterns.
Radiologist Action: Strongly recommend executing deep neural restoration or wavelet-based denoising. If diagnostic ambiguities persist, evaluate sensor hardware.

Radiologist Review Block	AI System Validation Specs
Reviewing Radiologist Signature	**Architecture:** U-Net++ Nested UNet (best_model.pth) **Dice Score:** 0.9886 | **IoU:** 0.9778 **Classes:** Gaussian, Poisson Noise

Disclaimer: This report was automatically generated by Model 1 (U-Net++) for multi-noise classification and severity assessment in educational & research environments. Findings should be cross-referenced by a qualified medical professional prior to clinical intervention.